**Q1 - 25 June - Shift 1**

A uniform chain of 6 m length is placed on a table such that a part of its length is hanging over the edge of the table. The system is at rest. The co-efficient of static friction between the chain and the surface of the table is 0.5, the maximum length of the chain hanging from the table is ________m.

*Space for your notes:*

**Q2 - 25 June - Shift 1**

A force on an object of mass 100g is $(10\hat{i}+5\hat{j})$ N. The position of that object at t = 2s is $(a\hat{i}+b\hat{j})$ m after starting from rest. The value of $\frac{a}{b}$ will be ________

*Space for your notes:*

**Q3 - 25 June - Shift 2**

A disc with a flat small bottom beaker placed on it at a distance R from its center is revolving about an axis passing through the center and perpendicular to its plane with an angular velocity ω. The coefficient of static friction between the bottom of the beaker and the surface of the disc is μ. The beaker will revolve with the disc if :

(A) $R \leq \frac{\mu g}{2\omega^2}$ (B) $R \leq \frac{\mu g}{\omega^2}$

(C) $R \geq \frac{\mu g}{2\omega^2}$ (D) $R \geq \frac{\mu g}{\omega^2}$

*Space for your notes:*

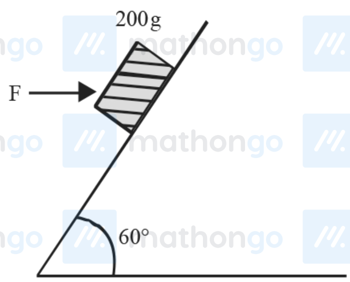
**Q4 - 25 June - Shift 2**

A curved in a level road has a radius 75m. The maximum speed of a car turning this curved road can be 30 m/s without skidding. If radius of curved road is changed to 48 m and the coefficient of friction between the tyres and the road remains same, then maximum allowed speed would be__ m/s.

*Space for your notes:*

**Q5 - 25 June - Shift 2**

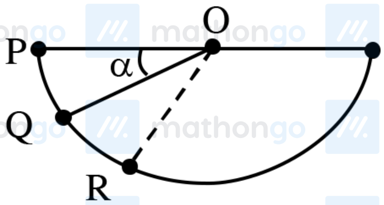
A block of mass 200 g is kept stationary on a smooth inclined plane by applying a minimum horizontal force $F=\sqrt{x}N$ as shown in figure. The value of $x=$ ________.

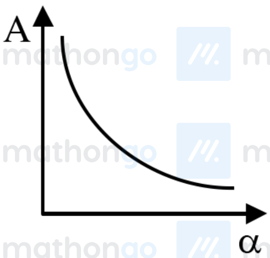


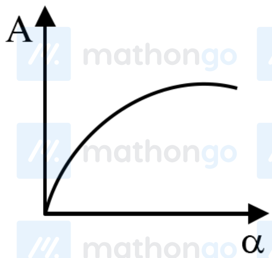

*Space for your notes:*

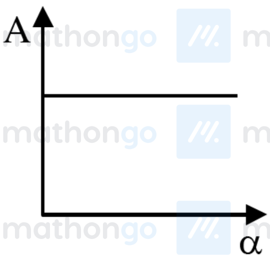
**Q6 - 26 June - Shift 1**

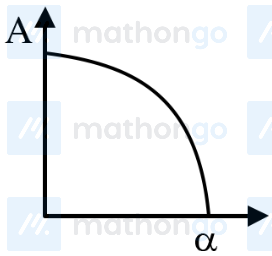
A ball is released from rest from point P of a smooth semi-spherical vessel as shown in figure. The ratio of the centripetal force and normal reaction on the ball at point Q is A while angular position of point Q is $\alpha$ with respect to point P. Which of the following graphs represent the correct relation between A and $\alpha$ when ball goes from Q to R ?




(A)



(B)


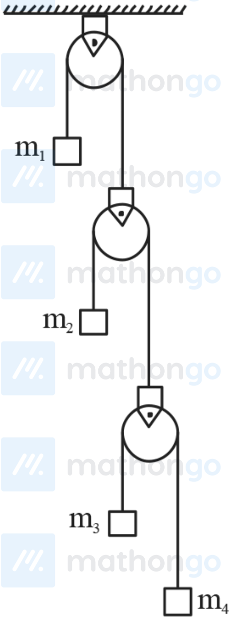

(C)



(D)



Q7 - 26 June - Shift 2

*Space for your notes:*

In the arrangement shown in figure $a_1, a_2, a_3$ and $a_4$ are the accelerations of masses $m_1, m_2, m_3$ and $m_4$ respectively. Which of the following relation is true for this arrangement?




(A) $4a_1 + 2a_2 + a_3 + a_4 = 0$

(B) $a_1 + 4a_2 + 3a_3 + a_4 = 0$

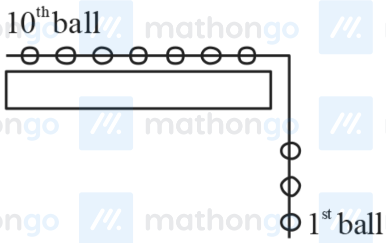
(C) $a_1 + 4a_2 + 3a_3 + 2a_4 = 0$

(D) $2a_1 + 2a_2 + 3a_3 + a_4 = 0$

*Space for your notes:*

Q8 - 26 June - Shift 2

A system to 10 balls each of mass 2 kg are connected via massless and unstretchable string. The system is allowed to slip over the edge of a smooth table as shown in figure. Tension on the string between the $7^{th}$ and $8^{th}$ ball is____________N when $6^{th}$ ball just leaves the table.

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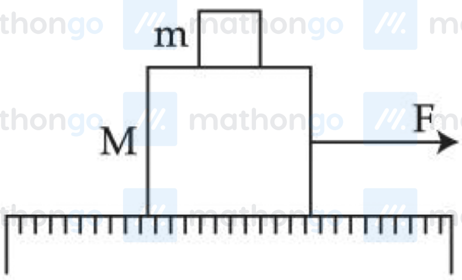
**Q9 - 27 June - Shift 1**

A system of two blocks of masses m = 2 kg and M = 8 kg is placed on a smooth table as shown in figure. The coefficient of static friction between two blocks is 0.5. The maximum horizontal force F that can be applied to the block of mass M so that the blocks move together will be :




(A) 9.8 N (B) 39.2 N

(C) 49 N (D) 78.4 N

*Space for your notes:*

**Q10 - 27 June - Shift 2**

One end of a massless spring of spring constant k and natural length $l_0$ is fixed while the other end is connected to a small object of mass m lying on a frictionless table. The spring remains horizontal on the table. If the object is made to rotate at an angular velocity ω about an axis passing through fixed end, then the elongation of the spring will be:

(A) $\frac{k - m\omega^2 l_0}{m\omega^2}$

(B) $\frac{m\omega^2 l_0}{k + m\omega^2}$

(C) $\frac{m\omega^2 l_0}{k - m\omega^2}$

(D) $\frac{k + m\omega^2 l_0}{m\omega^2}$

*Space for your notes:*

**Q11 - 27 June - Shift 2**

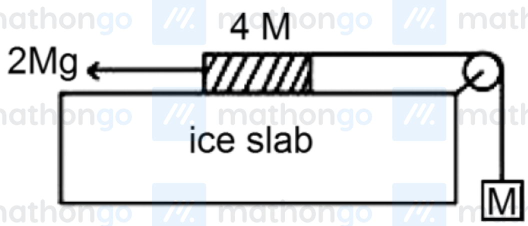
A mass of 10 kg is suspended vertically by a rope of length 5m from the roof. A force of 30 N is applied at the middle point of rope in horizontal direction. The angle made by upper half of the rope with vertical is $\theta = \tan^{-1}(x \times 10^{-1})$. The value of x is ______ .

(Given $g = 10 \text{ m/s}^2$)

*Space for your notes:*

**Q12 - 28 June - Shift 1**

A hanging mass M is connected to a four times bigger mass by using a string-pulley arrangement. as shown in the figure. The bigger mass is placed on a horizontal ice-slab and being pulled by 2 Mg force. In this situation. tension in the string is $\frac{x}{5}$ Mg for x = __________. Neglect mass of the string and friction of the block (bigger mass) with ice slab. (Given g = acceleration due to gravity)




*Space for your notes:*

**Q13 - 28 June - Shift 2**

A block of mass 2 kg moving on a horizontal surface with speed of 4 $ms^{-1}$ enters a rough surface ranging from x = 0.5 m to x = 1.5 m. The retarding force in this range of rough surface is related to distance by $F = -kx$ where k = 12 $Nm^{-1}$. The speed of the block as it just crosses the rough surface will be:

(A) Zero (B) 1.5 $ms^{-1}$

(C) 2.0 $ms^{-1}$ (D) 2.5 $ms^{-1}$

*Space for your notes:*

**Q14 - 29 June - Shift 1**

A wire of length L is hanging from a fixed support. The length changes to $L_1$ and $L_2$ when masses 1kg and 2 kg are suspended respectively from its free end. Then the value of L is equal to :

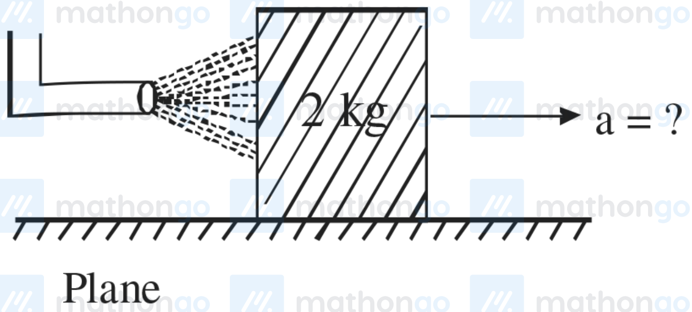
(A) $\sqrt{L_1 L_2}$ (B) $\frac{L_1 + L_2}{2}$

(C) $2L_1 - L_2$ (D) $3L_1 - 2L_2$

*Space for your notes:*

**Q15 - 29 June - Shift 1**

A block of metal weighing 2 kg is resting on a frictionless plane (as shown in figure). It is struck by a jet releasing water at a rate of 1 $kgs^{-1}$ and at a speed of 10 $ms^{-1}$. Then, the initial acceleration of the block, in $ms^{-2}$, will be :



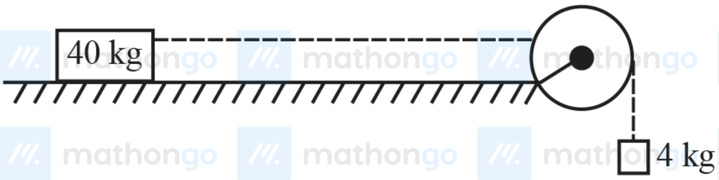

(A) 3 (B) 6

(C) 5 (D) 4

*Space for your notes:*

**Q16 - 29 June - Shift 2**

A block of mass 40 kg slides over a surface, when a mass of 4 kg is suspended through an inextensible massless string passing over frictionless pulley as shown below. The coefficient of kinetic friction between the surface and block is 0.02. The acceleration of block is. (Given $g = 10\ ms^{-2}$.)




(A) $1\ ms^{-2}$ (B) $1/5\ ms^{-2}$
(C) $4/5\ ms^{-2}$ (D) $8/11\ ms^{-2}$

*Space for your notes:*

**Q17 - 29 June - Shift 2**

A block of mass M placed inside a box descends vertically with acceleration 'a'. The block exerts a force equal to one-fourth of its weight on the floor of the box. The value of 'a' will be :

(A) $\frac{g}{4}$ (B) $\frac{g}{2}$ (C) $\frac{3g}{4}$ (D) g

*Space for your notes:*

# Answer Key

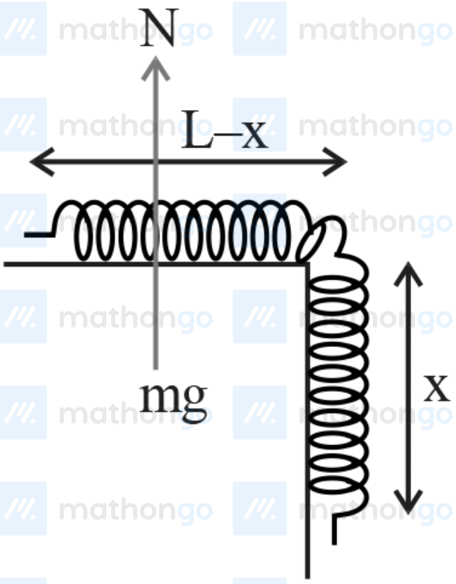
| | | | |
|---|---|---|---|
| **Q1** (2) | **Q2** (2) | **Q3** (B) | **Q4** (24) |
| **Q5** (12) | **Q6** (C) | **Q7** (A) | **Q8** (36) |
| **Q9** (C) | **Q10** (C) | **Q11** (3) | **Q12** (6) |
| **Q13** (C) | **Q14** (C) | **Q15** (C) | **Q16** (D) |
| **Q17** (C) | | | |

**Q1 (2)**

Mass per unit length $= \lambda$

$$N = mg = \lambda(L-x)g$$

$$fs_{max} = \mu_s N$$




$$fs_{max} = (0.5)(\lambda)(L-x)g$$

And also $fs_{max} = m_x g$

$$0.5\lambda(L-x)g = \lambda xg$$

$$\frac{L-x}{2} = x$$

$$\frac{L}{2} = \frac{3x}{2} \Rightarrow x = \frac{L}{3} = \frac{6}{3} = 2m$$

**Q2 (2)**

$$\vec{F} = 10\hat{i} + 5\hat{j}$$

$$m = 100 \text{ g} = 0.1 \text{ kg}$$

$$\vec{a} = \frac{\vec{F}}{m} = 100\hat{i} + 50\hat{j}$$

$$\vec{S} = \vec{u}t + \frac{1}{2}\vec{a}t^2 = \frac{1}{2}\vec{a}t^2 (\text{as } \vec{u} = 0)$$

$$= \frac{1}{2}(100\hat{i} + 50\hat{j})2^2$$

$$= 200\hat{i} + 100\hat{j}$$

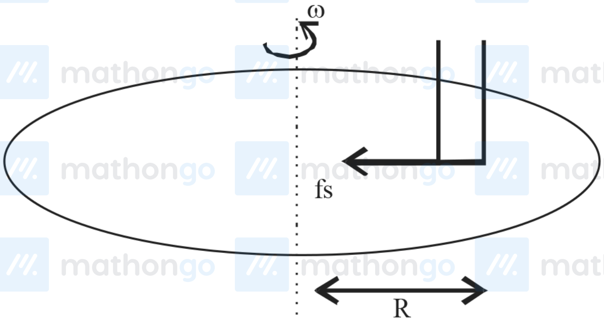
$$= a\hat{i} + b\hat{j}$$

$$a = 200, \ b = 100$$

$$\therefore \frac{a}{b} = 2$$

**Q3 (B)**

For beaker to move with disc




$f_s = m\omega^2 R$

We know that $f_S \leq f_{s\,max}$

$m\omega^2 R \leq \mu mg$

$R \leq \frac{\mu g}{\omega^2}$

**Q4 (24)**

$$f_{s\,max} = \frac{mv^2}{R}$$

$$\mu mg = \frac{mv^2}{R}$$

$$v = \sqrt{\mu Rg}$$

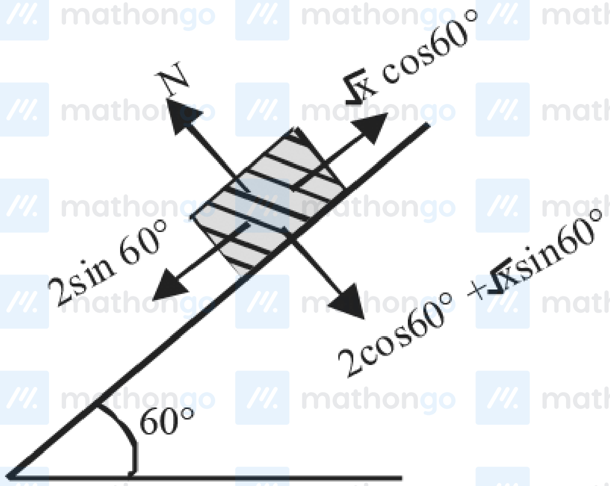
$$\frac{v_2}{v_1} = \sqrt{\frac{R_2}{R_1}}$$

$$\frac{v_2}{30} = \sqrt{\frac{48}{75}}$$

$$v_2 = 24\,m/s$$

**Q5 (12)**

$$mg = 2N$$



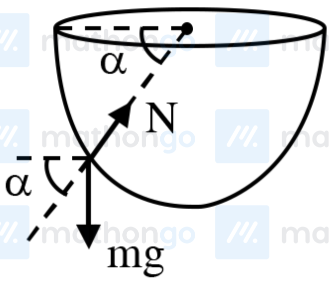

$$\sqrt{x}\frac{1}{2} = \frac{2\sqrt{3}}{2}$$

$$x = 12$$

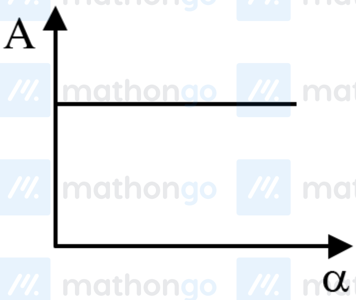
**Q6 (C)**

$V=\sqrt{2gR\sin\alpha}$

$$N - mg\sin\alpha = \frac{mv^2}{R} = 2mg\sin\alpha$$

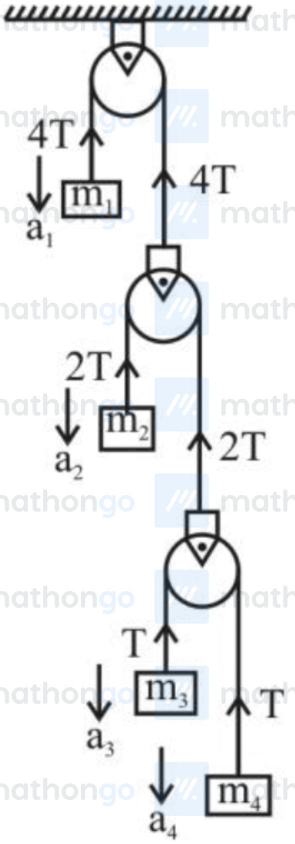



$$\frac{N}{2mg\sin\alpha} = \frac{1}{2} + 1 = \frac{3}{2}$$




$\Rightarrow A = \text{constant}$

**Q7 (A)**



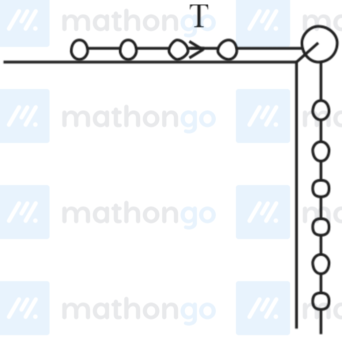

Using costraint

$\sum \vec{T}.\vec{a} = 0$

$-4Ta_1 - 2Ta_2 - Ta_3 - Ta_4 = 0$

$4a_1 + 2a_2 + a_3 + a_4 = 0$

**Q8 (36)**




$$a = \frac{6mg}{10m} = \frac{6g}{10} = \frac{3g}{5}$$

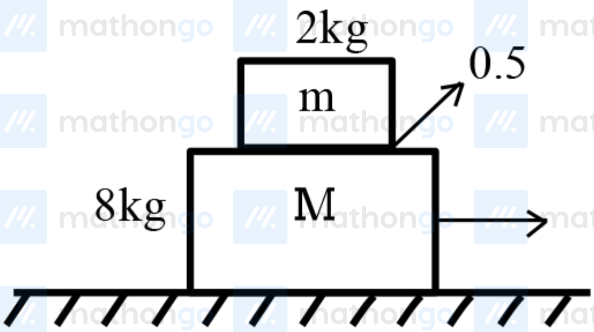
taking 8,9,10 together 


$T = 3\ ma$

$= 3m \times \frac{3g}{5}$

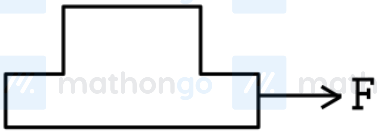
$= 36\ N$

**Q9 (C)**



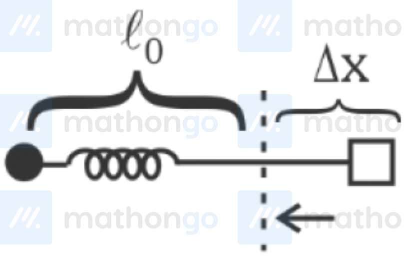

$$(a_A)_{\max} = 0.5g = 4.9\,m/s^2$$

For moving together




$$F_{\max} = m_T a_A$$
$$= 10 \times 4.9$$
$$= 49N$$

**Q10 (C)**

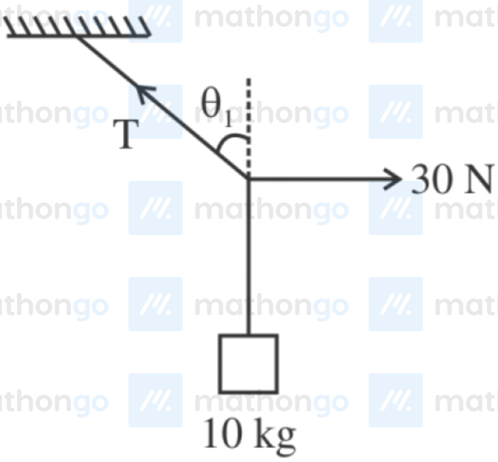



$$K\,\Delta x = m(\ell_0 + \Delta x)w^2$$

$$K\,\Delta x = m\,\ell_0\,w^2 + mw^2\,\Delta x$$

$$\Delta x = \frac{m\ell_0 w^2}{k - mw^2}$$

**Q11 (3)**




$$T \sin\theta = 30$$

$$T \cos\theta = 100$$

$$\Rightarrow \quad \tan\theta = 0.3$$

**Q12 (6)**

Using $\vec{F}_{net} = \mu\vec{a}$,

$$\begin{aligned} 2Mg - T &= 4Ma \\ T - Mg &= Ma \\ \hline \Rightarrow a &= \frac{g}{5} \end{aligned}$$

$$T = Mg + Ma = Mg + \frac{Mg}{5} = \frac{6}{5}Mg$$

**Q13 (C)**

$$a = \frac{-kx}{2} = \frac{-12x}{2} = -6x$$

$$\frac{vdv}{dx} = -6x$$

$$\int_4^v vdv = -\int_{\frac{1}{2}}^{3/2} 6xdx$$

$$\frac{v^2 - 4^2}{2} = -\frac{6}{2}\left[\left(\frac{3}{2}\right)^2 - \left(\frac{1}{2}\right)^2\right]$$

$$v^2 - 16 = -6\left(\frac{9}{4} - \frac{1}{4}\right)$$

$$v^2 = 16 - 6 \times 2 = 4$$

$$V = 2 \text{ m/s}$$

**Q14 (C)**

By Hooke's Law

so $F \alpha \Delta L$

$$\frac{F_1}{F_2} = \frac{\Delta L_1}{\Delta L_2}$$

$$\frac{10}{20} = \frac{(L_1 - L)}{(L_2 - L)}$$

$$L = 2L_1 - L_2$$

**Q15 (C)**

$$F = \frac{dp}{dt} = \upsilon\frac{dm}{dt}$$

$\Rightarrow Ma = 10 \times 1$

$\Rightarrow 2a = 10$

$a = 5 m/sec^2$

**Q16 (D)**

For 4 kg block

$4g - T = 4a$

For 40 kg block

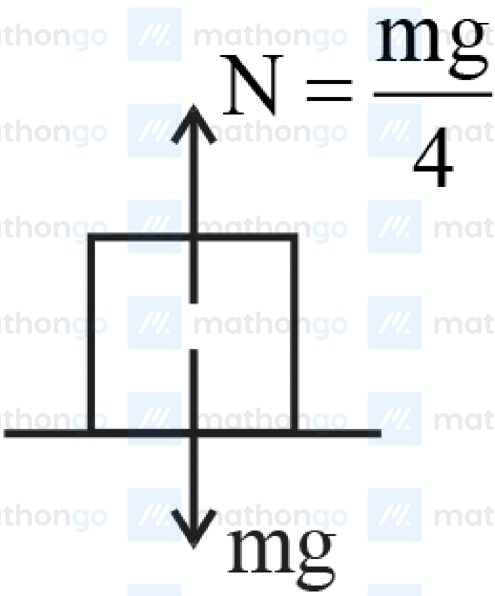
$T - 40g \times 0.02 = 40\ a$

Adding both eq.

$40 - 8 = 44a$

$$a = \frac{32}{44} = \frac{8}{11} m/s^2$$

**Q17 (C)**

- 

$$mg - N = ma$$

$$a = g - \frac{g}{4}$$

$$a = \frac{3g}{4}$$